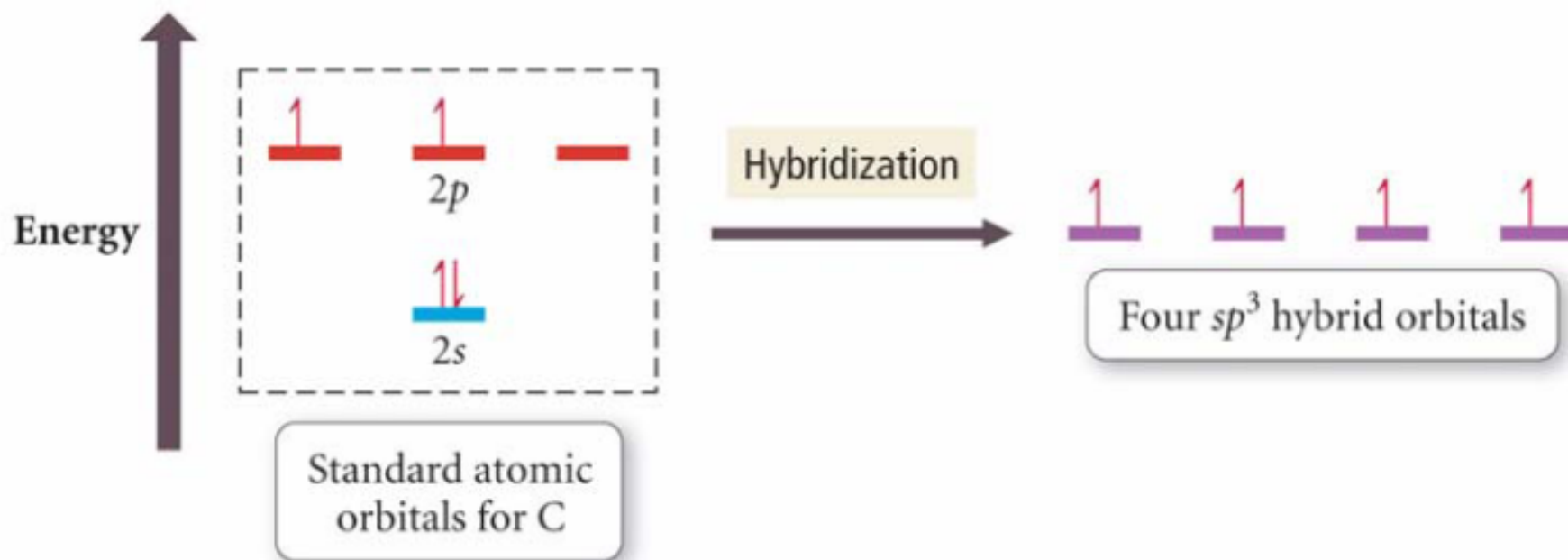


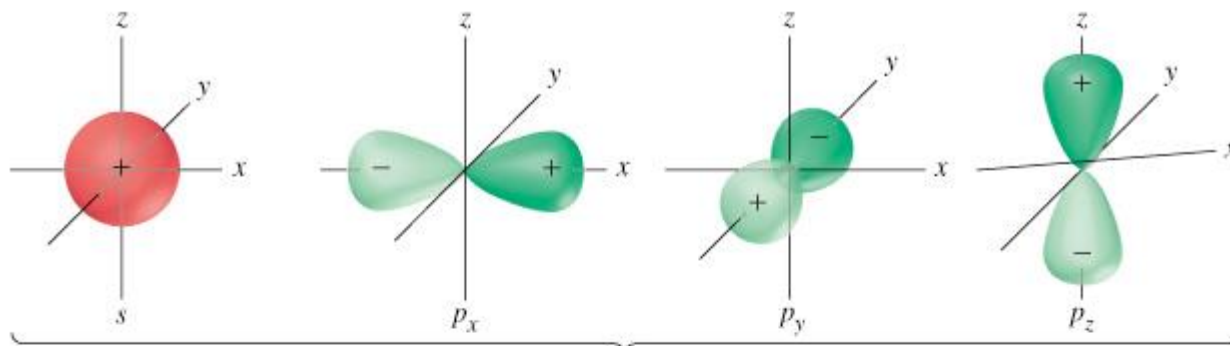
sp³ Hybridization

- When the *s* and all three *p* orbitals combine, the resulting hybrid orbitals are ***sp³ hybrid orbitals***.

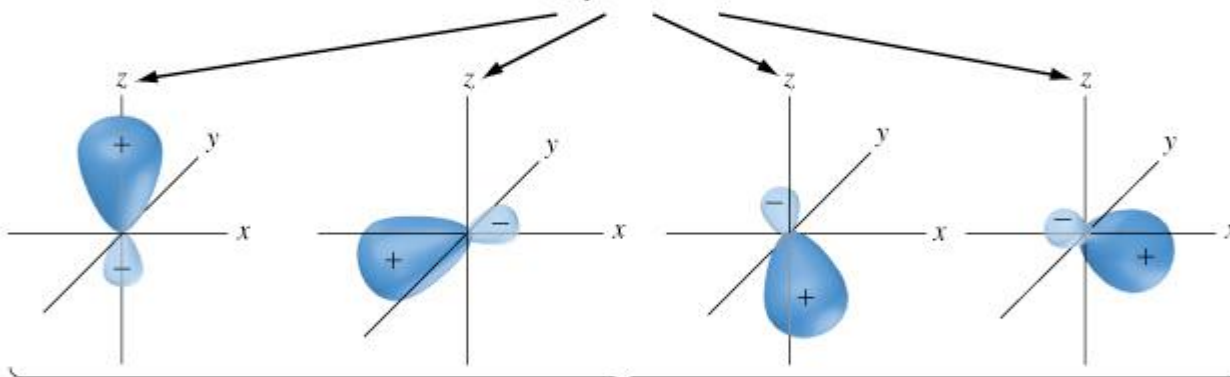


- These 4 orbitals point to the corners of a tetrahedron (109.5°).
- sp^3 orbitals are used to describe the C atoms in CH_4 and C_2H_6 , the N in NH_3 , and the O in H_2O .

sp^3 Hybridization



Combine to generate
four sp^3 orbitals

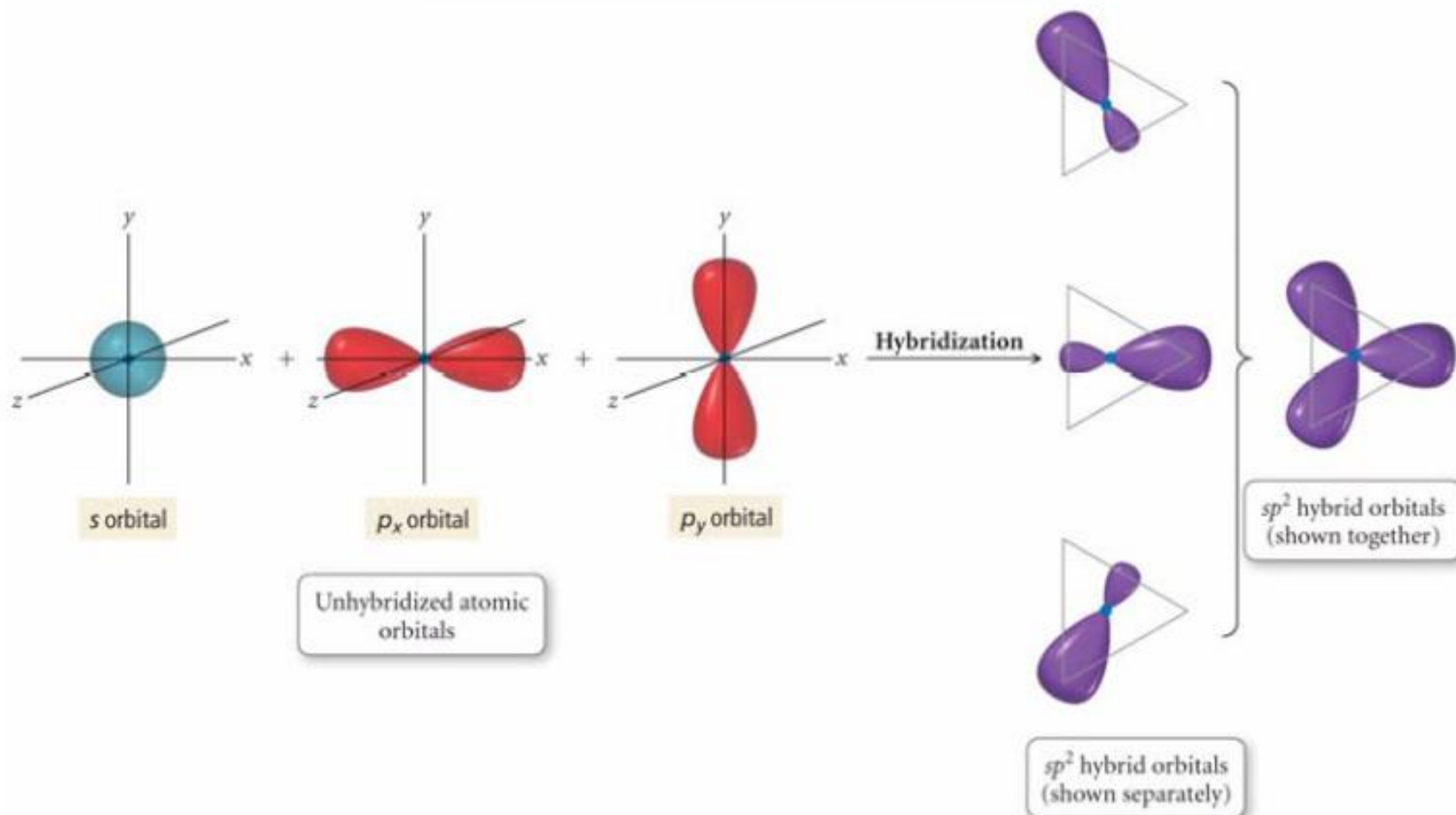


Which are represented
as the set



sp^2 Hybridization

One s orbital and two p orbitals combine to form three sp^2 orbitals.



sp^3d and sp^3d^2 Hybridization

- The s and the three p orbitals can also combine with d orbitals in molecules with expanded octets:

– $s + p + p + p + d = \mathbf{sp^3d}$ hybrid orbitals

– $s + p + p + p + d + d = \mathbf{sp^3d^2}$ hybrid orbitals

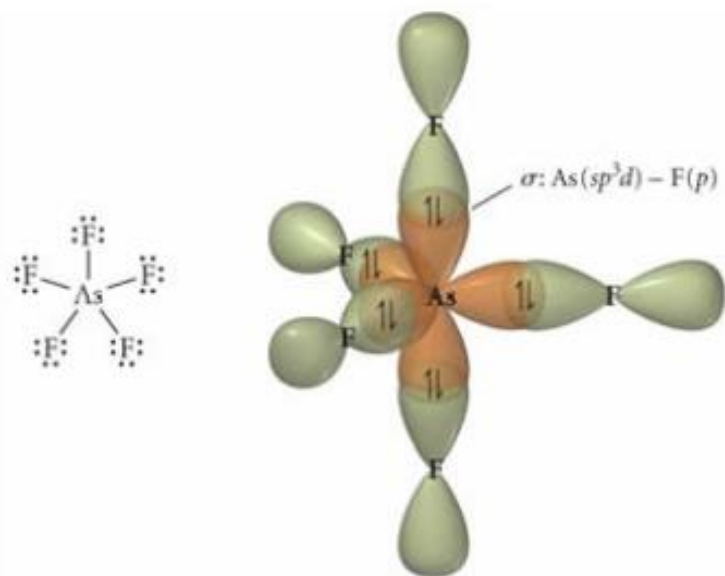


sp^3d hybrid orbitals
(shown together)

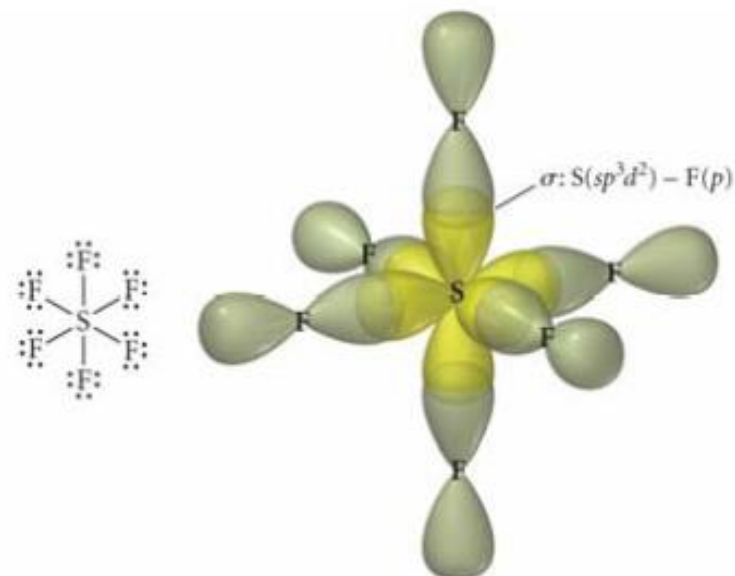


sp^3d^2 hybrid orbitals
(shown together)

sp^3d Orbitals in AsF_5 ; sp^3d^2 Orbitals in SF_6



sp^3d hybrid orbitals in AsF_5



sp^3d^2 hybrid orbitals in SF_6

Hybrid Orbitals

2 charge clouds $\rightarrow$ 2 sp orbitals $\rightarrow$ linear (180°)

3 charge clouds $\rightarrow$ 3 sp^2 orbitals $\rightarrow$ trigonal planar (120°)

4 charge clouds $\rightarrow$ 4 sp^3 orbitals $\rightarrow$ tetrahedral (109.5°)

5 charge clouds $\rightarrow$ 5 sp^3d orbitals $\rightarrow$ trigonal bipyramidal ($90^\circ, 120^\circ$)

6 charge clouds $\rightarrow$ 6 sp^3d^2 orbitals $\rightarrow$ octahedral (90°)